

Facility for calibration of optical and condensation particle counters based on a turbulent aerosol mixing tube and a reference optical particle counter

Cite as: Rev. Sci. Instrum. **90**, 075111 (2019); <https://doi.org/10.1063/1.5095853>

Submitted: 13 March 2019 . Accepted: 18 June 2019 . Published Online: 15 July 2019

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ABSTRACT

In this study, we present a custom-made facility developed for evaluating the measurement efficiency of optical particle counters and other common aerosol instrumentation. The facility consists of an aerosol generation setup, a turbulent flow tube for particle homogenization, isokinetic sampling ports, and a home-built particle counter that serves as a reference instrument. Stable and reproducible aerosols of polystyrene latex particles can be produced in the size range 100 nm–10 μm and at concentrations between 0.5 cm^{-3} and a few thousand submicron particles per cm^3 or a few tens of 10 μm particles per cm^3 . The flow characteristics in the homogenizer were investigated with laser Doppler velocimetry measurements and computational fluid dynamics simulations indicating a plug (turbulent) flow at the aerosol sampling location. The particle mixing characteristics were determined experimentally at various heights of the flow tube by parallel measurements with two condensation particle counters. A spatial homogeneity within 1.1% was found across the sampling area. The measurement uncertainties in the determination of the particle number concentration have been evaluated in detail and amount typically to 5.4% at 1 cm^{-3} and 2.1% at 100 cm^{-3} (95% confidence level).

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I. INTRODUCTION

Optical particle counters (OPCs) are increasingly used in cleanrooms (e.g., in semiconductor industries and medical and pharmaceutical environments) for monitoring the number concentration of airborne particles. A cleanroom is an enclosed and highly controlled space in which airborne particles, contaminants, and pollutants are kept within strict limits. In industry, cleanrooms are used in manufacturing and servicing of hardware, such as integrated circuits and hard drives. In biotechnology and pharmaceutical sciences, cleanrooms are used to minimize the presence of bacteria, viruses, and other pathogens. Without strict control, contamination can compromise severely the quality of the (manufactured) products or processes, lead to disease spreading in hospital environments,^{1,2} or aggravate the health of industry workers.^{3–5} Hence, the

development of accurate calibration procedures for OPCs is highly important.

In the past, the OPC measurement efficiency was evaluated by collecting and counting particles on filters^{6,7} or a silicon wafer.⁸ In recent years, the requirements for OPCs used in cleanrooms (also known as laser scattering airborne particle counters) have been defined in the ISO 21501-4 international standard.⁹ According to this standard, the counting efficiency of the devices under test (hereafter referred to as DUT-OPCs) must be evaluated with polystyrene latex (PSL) spheres of known size. In practice, the DUT-OPC samples during calibration a well-homogenized aerosol of PSL spheres at low concentrations, typically a few particles per cm^3 . Parallel measurements with a reference instrument, either a calibrated condensation particle counter (CPC)^{10–12} or a reference OPC,¹³ are carried out in order to determine the counting efficiency of the

DUT-CPC. In industry, the use of a CPC as a reference instrument for OPC calibration is very common even though the method suffers from several disadvantages. Traceability for CPCs has been established through comparison with electrometers only for number concentrations above 1000 cm^{-3} .¹⁴ Below this limit, a proportionality assessment over the range of the CPC must be carried out.¹⁵ Moreover, CPCs must be combined with a Differential Mobility Analyzer (DMA) for upstream particle size classification and are therefore only suited for particle sizes up to $1\text{ }\mu\text{m}$,^{10–12} with the exception of large custom-made DMA columns as reported by Järvinen *et al.*¹⁶ As a result, commercial DMA-CPC setups cannot respond to the needs of the medical/pharmaceutical industries, for which monitoring of large particles (up to $10\text{ }\mu\text{m}$ or larger) is essential.

Recently, Iida and Sakurai¹⁷ demonstrated a new method for evaluating the counting efficiencies of optical particle counters based on an inkjet aerosol generator (IAG).^{17,18} IAG generates particles at a constant rate, eliminating thus the need for a reference optical particle counter, a remarkable feature of the method. The method is well suited for generating NaCl (sodium chloride) and lactose monohydrate particles with aerodynamic and volume equivalent diameters up to several micrometers. However, the smallest particle size that can be generated is limited by the size of the residue particles arising from solvent (water) impurities.^{17,18} In practice, generating particles with d_{PSL} (i.e., diameter of PSL spheres with light-scattering intensity equal to that of a given test particle) below 500 nm can be challenging. Typically, the aerosol flow rate in the IAG is set to 0.3 l/min and the particle generation rate is set to 30 s^{-1} .¹⁷ Since the sampling flow rate of most commercially available DUT-OPCs is larger, the mismatch of the flowrates is compensated by adding particle-free air into the flow chamber of the experimental setup.¹⁷ This implies that the higher the flow rate of the DUT-OPC, the lower the sampled particle concentration will be, with number concentrations typically well below 1 cm^{-3} . Measurement uncertainties remain, however, small even at such low concentrations, thanks to the regular injection pattern, which ensures that uncertainties due to random particle distribution remain negligible.

In this study, we present a custom-made facility, which was developed primarily for determining the measurement efficiency of OPCs for cleanroom monitoring according to ISO 21501-4.⁹ Additional applications include calibration of CPCs (especially in the range $<1000\text{ cm}^{-3}$), Light-Scattering Aerosol Spectrometers (LSAs, also known as Optical Particle Size Spectrometers OPSSs) according to ISO 21501-1¹⁹ as well as Aerodynamic Particle Sizers (APSs).²⁰

The design of the facility is based on an early study by Schlatter in 2009²¹ but has been considerably improved to extend the measurement capabilities. It comprises an aerosol generation setup, a vertical flow tube for particle homogenization and a reference particle counter. The method described here is complementary to the IAG method¹⁷ and extends the measurement capabilities to both higher number concentrations and smaller particle sizes. More specifically, our method relies on the use of PSL particles of certified size in the range 100 nm – $10\text{ }\mu\text{m}$ and is capable of generating particle number concentrations between 0.5 cm^{-3} and several hundred/thousand particles cm^{-3} .

The first section of this article provides a detailed description of the aerosol generation and homogenization setup, the isokinetic sampling ports, and the reference particle counter. Section III focuses on the study of the flow characteristics performed with Laser Doppler Velocimetry (LDV) measurements and Computational Fluid Dynamics (CFD) simulations. Finally, the particle homogenization characteristics as measured with two CPCs, a thorough uncertainty budget, and an example of an OPC calibration are presented.

II. EXPERIMENTAL SETUP

As depicted in Fig. 1, the experimental setup consists of three distinct sections: (a) the aerosol generation system, (b) a 4-m-long turbulent flow tube (homogenizer), and (c) the particle detection system, which are described in detail below.

A. Aerosol generation setup

Wet or dry dispersion processes transfer the PSL particles (supplied by Thermo Scientific, Duke Scientific Corp., JSR Corp., Sigma-Aldrich, or Microbeads AS) in the aerosol phase. Depending on the desired particle size, three ranges (100 nm – 250 nm , 300 nm – 1000 nm , and $>1000\text{ nm}$) can be distinguished, each one requiring a different aerosolization procedure as described below.

- (i) Aqueous suspensions of PSL particles with a geometric diameter between 100 nm and 250 nm are aerosolized using a pneumatic Meinhard nebulizer equipped with a cyclone to remove large size droplets, which might contain more than one PSL particles. The aerosol is then guided through a dryer, i.e., a glass column filled with silica gel particles, and a DMA (Model 3071 A, TSI, Inc.) operated under a constant voltage. The latter has been preselected in a way that only PSL particles of the desired size enter the homogenizer, while residue particles (typically $<90\text{ nm}$ diameter) are removed. The aerosol is then delivered into the flow tube homogenizer in the downward direction. The concentration of the airborne PSL particles can be tuned by adjusting the concentration of the aqueous PSL suspensions accordingly. In general, high dilution factors are needed to avoid generating droplets containing multiple PSL particles and for minimizing the coating around the PSL particles that results from additives after droplet evaporation. It must be noted here that with the Meinhard nebulizer, particles with diameter up to $1\text{ }\mu\text{m}$ can be aerosolized, but for particles larger than 300 nm , the more straightforward nebulization procedure described below [case (ii)] is often preferred.
- (ii) PSL particles in the range 300 nm – $1\text{ }\mu\text{m}$ are generated from an aqueous suspension using an air-assisted nebulizer (model T1H ARIDUS, Teledyne CETAC Technologies) installed inside the homogenizer. The air flow is typically set to 1 l/min . The nebulizer is equipped with a custom-made impinger in order to break up the large water droplets. As a result, the size of the residue particles becomes much smaller and well separated from the peak of the PSL particles so that no size selection with a DMA is necessary for calibration of

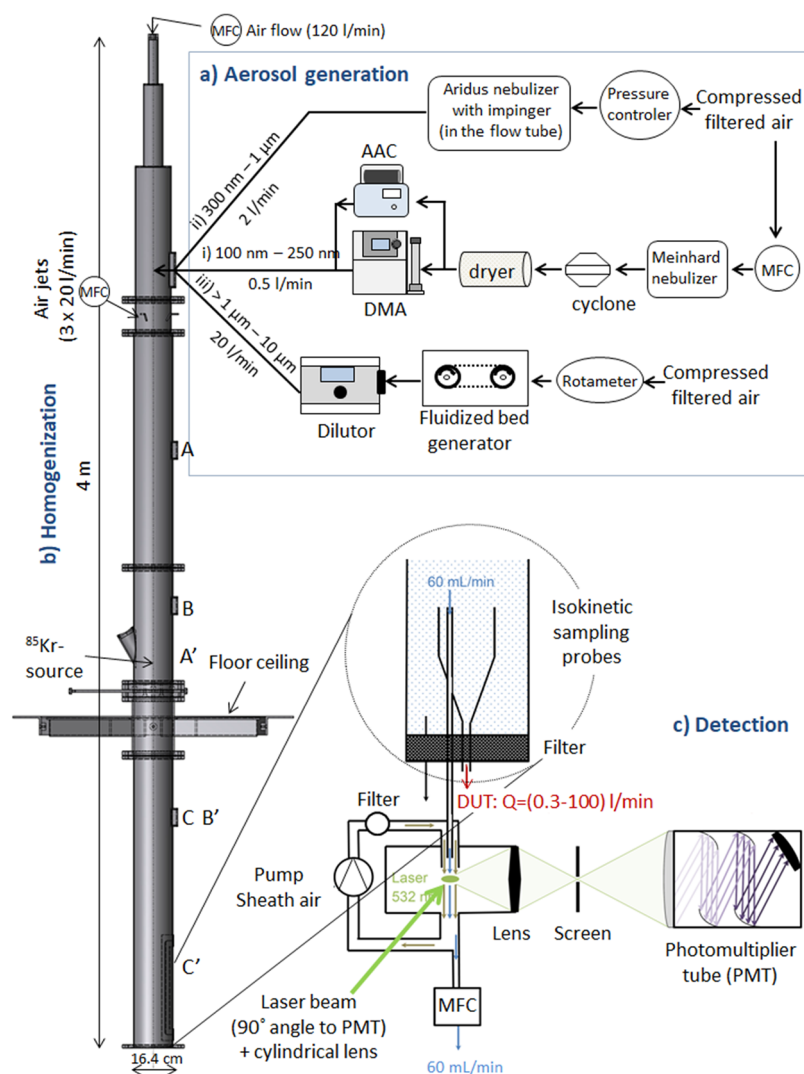


FIG. 1. Schematic representation of the experimental setup consisting of (a) the aerosol generation system, (b) a 4-m-long flow tube (homogenizer), and (c) the particle detection system. FC and MFC stand for flow controller and mass flow controller, respectively. Particle velocity measurements were carried out at positions A, B, and C corresponding to a distance of 0.75, 1.4, and 2.4 m from the aerosol injection point, respectively. Concentration measurements were carried out at positions A', B', and C', corresponding to 1.7, 2.4, and 3.0 m downstream of the aerosol injection point.

OPCs. Additionally, the probability of two particles being present in one droplet is considerably reduced.

- (iii) PSL particles larger than $1\ \mu\text{m}$ are generated with a dry process using a fluidized bed generator (Model 3400, TSI, Inc.). These enter the homogenizer right after a dilution step (Model HDS 561 dilution unit, Topas) through an injection pipe (20 l/min total flow) pointing downward.

For calibration of instruments such as CPCs, which do not classify particles according to their size, a DMA or Aerodynamic Aerosol Classifier (AAC, Cambustion)^{22,23} is needed to select the peak of the PSL particles and filter out the undesired residue particles before injecting the aerosol into the homogenizer. In this case, particles in the size range 100 nm– $1\ \mu\text{m}$ are generated with the Meinhard nebulizer as described in case (i). Due to the low particle transmission efficiency of the DMA (only 30% of the 100 nm particles and 10% of the $1\ \mu\text{m}$ particles get a single charge in the bipolar charger²⁴), the concentration of PSL particles after dilution in the

homogenizer is relatively low. Instead, if an AAC is used (transmission efficiency of about 80% in this size range), higher number concentrations by about a factor of 4 can be achieved.²³ The AAC is also suitable for classifying particles larger than $1\ \mu\text{m}$ and can be combined with the dry aerosolization process [case (iii)]. For OPCs and other instruments that can measure particle size, particle size selection is usually not necessary. Table I summarizes the particle number concentration ranges which can be attained with the three aerosol generation systems described above with and without particle size selection.

B. Aerosol flow-tube homogenizer

The homogenizer is a 4-m-long custom-made stainless steel tube with an inner diameter of 16.4 cm and is placed vertically, extending over two floors through an opening in the ceiling. As shown in Fig. 1, the aerosol generation system is located in the upper floor. Dry filtered air with a fixed flow rate of 120 l/min enters the

TABLE I. Aerosol generation methods for different particle sizes, approximate PSL particle number concentrations with and without size classification, and application field.

Aerosol generation method	Particle size (μm)	Concentration range (cm^{-3})	Suitable for CPC calibration	Suitable for OPC/LSAS/APS calibration
(i) in combination with AAC (DMA)	0.1–0.5	0.5–800(200)	Yes	Yes
(i) in combination with AAC (DMA)	>0.5–1.0	0.5–400(100)	Yes	Yes
(ii) no size selection	0.3–1.0	0.5–2000	No	Yes
(iii) no size selection	>1.0–10.0	0.5–50	No	Yes
(iii) in combination with AAC	>1.0–>5.0	0.5–20	Yes	Yes

homogenizer from the very top and sweeps the PSL particles down the tube, where they are further mixed by three turbulent jets of air. The air-jet injection tubes are placed symmetrically around the homogenizer tube pointing 60° downwards. The flow rate of the air-jets is adjusted so that the final (total) aerosol flow is equal to 180 l/min. As the aerosol travels down the flow tube, the water surrounding the PSL particles (in case of wet dispersion) is fully vaporized and the particles are completely dry when they reach the sampling zone. To reduce particle losses due to electrostatic forces, a ^{85}Kr -radioactive source of ≈ 1 MBq activity is used as a “neutralizer” (i.e., ionizes air molecules, which then attach to charged PSL particles so that the aerosol becomes overall neutral).

The sampling zone (marked C' in Fig. 1) is located 3.0 m downstream of the injection position and accommodates two isokinetic sample probes placed at the bottom end of the homogenizer. Isokinetic conditions are necessary when sampling with two optical particle counters operating at different flow rates to ensure representative sampling and minimize sampling artifacts of larger particles. Several custom-made sampling probes with a narrowing angle of 34° but different cross sections have been therefore designed to match the flow rate of the various DUT-OPCs, which typically ranges between 0.3 l/min and 100 l/min.

The sampling probe of the reference particle counter is also a probe funnel with size adjusted to the air velocity and a narrowing angle, again, of 34° . More specifically, this sampling probe is a pipe with an inner diameter of 3.0 mm, which narrows gradually down to a diameter of 0.2 mm just upstream of the optical detection unit. The “particle loss calculator”²⁵ was used to validate this configuration. According to the calculations, impaction losses in the probe are negligible, while diffusion losses are about 1% for 500 nm particles, 0.5% for 1 μm particles, and 0.1% for 10 μm particles. These losses are corrected for during data treatment (see Table II). It is worth mentioning that the sampling probe of the reference counter is placed within the sampling probe of the DUT slightly off axis. This design ensures that both instruments sample aerosol from the central cross section of the flow tube, where the aerosol is found to be homogeneous within 1.1% (see Subsection IV B). The velocity profile at the sampling area, as will be presented in Sec. IV A, is nearly uniform.

C. Reference optical particle counter

The detection system consists of a custom-built optical particle counter placed right at the outlet of the homogenizer to

minimize particle losses due to tube bendings. The sampled aerosol enters the detection chamber through the 0.2 mm orifice of the sampling probe (aerosol direction is designated by a blue arrow in Fig. 1) and is surrounded by a sheath-air flow (3.5 l/min, brown arrows) which prevents the particle beam from diverging. The sheath air is recirculated and filtered; hence, the sampling flow can be determined by measuring the outflow only. A laser beam of 0.7 mm width is generated by a continuous-wave laser (Model Verdi V-5, Coherent) at a wavelength of 532 nm and focused at the point of intersection with the aerosol stream with the use of a cylindrical lens. Particles crossing the laser beam scatter light, which is detected by a photomultiplier tube placed at a 90° angle.

Before the measurements start, the laser beam needs to be carefully aligned through the optical cell to ensure that the aerosol beam/jet exiting the 0.2 mm nozzle crosses the laser beam (0.7 mm width) approximately at the center and consequently, all aerosol particles are detected by light scattering. It should be noted that despite a possible jet divergence, its width remains much smaller than that of the laser beam. Imperfect alignment of the laser beam and the aerosol jet results in asymmetric intensity distributions of time integrated scattered light, exhibiting a tail toward low integrated intensity (appearing as smaller particle sizes). This occurs due to the Gaussian intensity distribution of the laser beam. The alignment is therefore repeated carefully until this intensity distribution becomes symmetric.

TABLE II. Table summarizing the physical quantities and their measurement uncertainties required for the evaluation of the reference aerosol particle number concentration in the case of 500 nm PSL particles at number concentration 1.0 cm^{-3} .

Quantity	Value (example)	Standard uncertainty	Relative uncertainty
N_{average}	60.00	1.56	5.2% (95% conf. level)
η_{noise}	0	Negligible	Negligible
Q	60.00 ml/min	0.09 ml/min	0.3% (95% conf. level)
P	0.99		1% ²⁷
t	60 s	Negligible	Negligible
η_{hom}	1.0000	0.0045	1.1% (95% conf. level)
c_{ref}	1.008 cm^{-3}	0.027 cm^{-3}	5.4% (95% conf. level)

III. EXPERIMENTAL AND NUMERIC TECHNIQUES FOR FLOW INVESTIGATION

A. Laser Doppler velocimetry (LDV) technique

To characterize the flow profile, a LDV system (Model fp50shift-532 nm, ILA) with a 160 mm lens was used. The principle of this instrument is based on two crossed laser beams, which give rise to a fringe pattern. If a particle crosses this volume (diameter and length of approx. 100 μm and 0.5 mm, respectively), it scatters light modulated with a frequency due to that pattern and is detected by a photo multiplier. From this frequency, the particle velocity can be derived. According to the manufacturer's data sheet, the instrument can measure velocities down to 0.01 m/s at an accuracy of 0.3%. For each measurement position, 1000 validated velocity signals were recorded.

The Reynolds (Re) number of the flow, $Re = \frac{\rho u L}{\mu}$, where ρ and u designate the density and velocity of the fluid, respectively, L is the diameter of the flow tube, and μ is the dynamic viscosity, is calculated to be 1800 assuming a plug flow profile. DEHS (di-ethyl-hexyl-sebacate) fog was injected (10 l/min) through a 10 mm inner diameter pipe at the position where the PSL particles are usually injected. This standard procedure created a sufficient amount of seed particles to achieve data rates of 10–100 Hz, with the lower value occurring toward the wall. For this purpose, the openings marked as A, B, and C in Fig. 1 were used and covered by a 22 cm^2 glass window. Since the distance of the two laser beams was 45 mm when exiting the lens, it did not allow us to measure a full cross-stream profile in the flow tube.

B. Computational fluid dynamics (CFD) simulations

CFD calculations were performed using the COMSOL Multi-physics package. Different mesh types were tested. COMSOL automatically generates grids consisting of tetrahedrons, which can be refined iteratively based on the calculated solution. Due to the long flow tube used in the current geometry, however, the automatic grid generation led to an unnecessarily high number of computational cells. After the initial mixing of the jets, the gradients in the stream-wise direction are very small. Therefore, a swept hexahedral grid was used, instead, starting from 0.3 m downstream of the aerosol injection position, as shown in Fig. 2(a).

Reynolds-averaged modeling of the Navier-Stokes equations was used. This method separates the velocity in a mean and fluctuating part [the latter is characterized by the root mean square (rms)]. The vector velocity, inserted in the Navier-Stokes equations, leads to higher-order correlations between velocity components, which need to be modeled. To this end, two k-epsilon models were tested: (i) the standard formulation including wall functions so that it is not necessary to resolve the flow with a fine grid close to the walls and (ii) a low-Reynolds number formulation, which resolves the boundary layer flow at the wall (for details, see Ref. 26). In the wall region, the model also used adjustments of the k- and epsilon-model equations to capture viscosity effects which become important close to the wall.

Initial tests with the standard k-epsilon model, including wall functions to describe the flow close to the wall, failed to reproduce the flow in the flow tube section. Instead, the low-Reynolds number k-epsilon turbulence model with a resolved (mesh-refined)

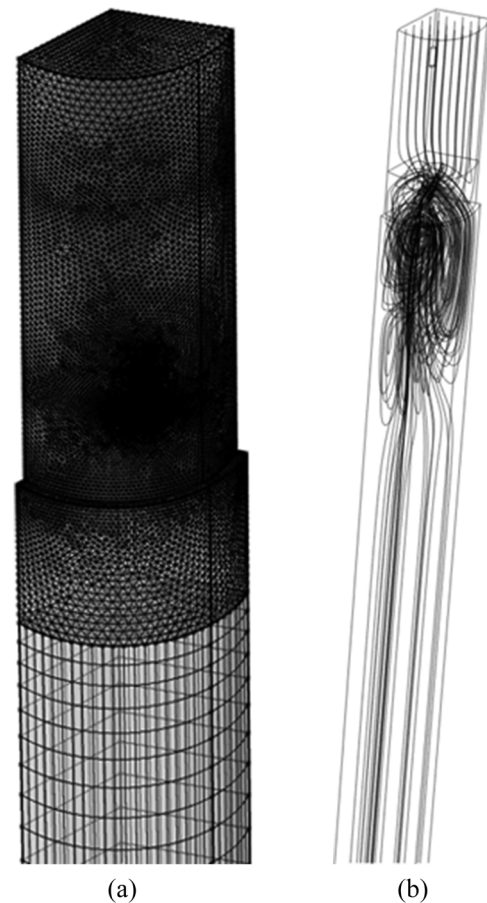


FIG. 2. (a) Mesh for the numeric simulation and (b) resulting streamlines. In (a), an increase of 10 mm in the tube diameter right after the aerosol injection point is visible. This is due to the combination of two slightly different standard vertical steel tubes (DIN EN 10357 and EN ISO 1127) and has a negligible effect on the flow profile.

boundary layer was used, which was capable of calculating the developing turbulent flow structure satisfactorily. However, in the jets/mixing region, this model requires a very detailed mesh, which was not compatible with the available computational resources. A rather coarse mesh was used in the jet region, which led to an overestimation of velocity gradients and turbulent kinetic energy as discussed in Subsections IV A and IV B.

The particles were modeled as a passive transported species with a diffusion coefficient according to their size. Lagrangian particle tracing was also tested but abandoned because of the high computational time arising from the small particle relaxation times and, hence, small computational time steps.

C. Particle concentration measurements

To characterize the homogeneity of the PSL particles in the flow tube (positions B' and C', see Fig. 1) as a function of particle size, PSL particles with a diameter of (500 ± 12) nm and (3000 ± 38) nm were

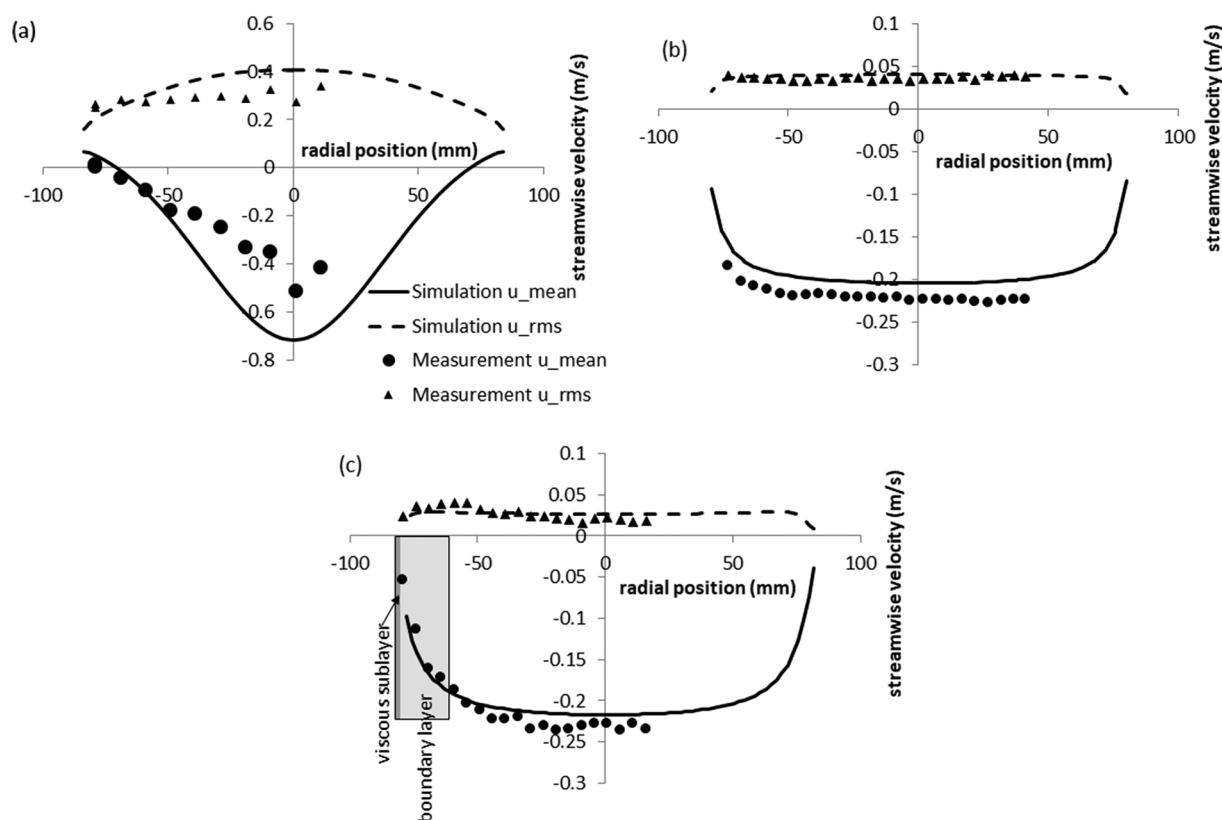


FIG. 3. Measured and calculated mean and rms velocities at downstream positions A, B, and C (see Fig. 1) corresponding to (a) 0.75 m, (b) 1.7 m, and (c) 2.4 m distance from the aerosol injection point, respectively.

generated using the aerosol generation systems (i) and (iii), respectively, as described in Sec. II A. For simplicity, we will refer to these particles as 500 nm and 3 μm PSL particles since the uncertainty in the size determination does not alter in any way the results of this study. Residues were removed with a DMA and an AAC. The resulting particle number concentration after homogenization was around 5 cm^{-3} .

Two parallel sampling lines of adjustable height were inserted into the flow tube; the position of the first sampling line was kept fixed at the center (radial position 0, see Fig. 3), whereas the other one was placed consecutively at a distance of -70 mm , -50 mm , $\dots$, $+70\text{ mm}$ with respect to the center. The outlet of each sampling line was connected to a calibrated CPC (Model 3775 and 3776, respectively, TSI, Inc.), and concentration measurements at eight different positions along the diameter of the flow tube were thus performed. During data analysis, the particle number concentration measured at the sampling line located at the center was used as reference ($C_{\text{reference}}$). The flow rate of each CPC was 0.3 l/min, and the inner diameter of the sampling line (tube) was 6 mm. This configuration ensured nearly isokinetic sampling.

With the standard CPC flow rate of 0.3 l/min and measurement times of 2 min, this resulted in measurement uncertainties of around $\pm 6\%$ – 10% , dominated by statistical uncertainties associated with the Poisson distribution (95% confidence interval) and

statistical uncertainties arising from instabilities in the generation of 3 μm particles (due to the use of the fluidized bed generator, see discussion in Subsection V A). To attain higher particle number concentrations and, hence, reduce measurement uncertainties, we generated DEHS aerosol (as used for LDV measurements, but diluted by a factor of 40) at concentrations of 5000 cm^{-3} . The droplet sizes ranged between 150 and 1500 nm with the median droplet size lying close to 400 nm. Measurements with the DEHS aerosol were performed at positions A', B', and C' (see Fig. 1) to investigate the evolution of mixing characteristics as a function of the flow tube length. The results are presented in Fig. 3 and discussed in Subsection IV B.

IV. RESULTS AND DISCUSSION

A. Velocity characteristics

Figure 2(b) presents the streamlines at the position of aerosol and mixing air injection and shows a relevant recirculation zone due to the mixing jets with a length of approximately 20 cm. After this length, the streamlines become parallel again. Figure 3 presents LDV measurements and CFD simulations of the mean particle velocity and the root mean square (rms) of the velocity fluctuations in time at three different heights of the flow tube

[positions A (0.75 m downstream of aerosol injection), B (1.4 m), and C (2.4 m) (see Fig. 1)]. It can be seen that the characteristics of the mean velocity and its fluctuations were captured well in the simulations for all three positions downstream of the mixing jets. Due to lack of optical access, however, it was not possible to measure the flow structure just downstream of the mixing jets.

By comparing panels (b) and (c) of Fig. 3, it can be seen that the boundary layer thickness increases with downstream position, the boundary layer being defined as the distance from the wall to the point where velocity reaches 90% of its centerline (i.e., maximum) value. It should be noted here that LDV measurements cannot extend close to the wall since there were hardly any DEHS particles present. Additionally, disturbances of the signal due to laser light reflection at the wall may occur. The structure of the boundary layer is of importance for the performance of the facility since it is the key mechanism for particle wall deposition and particle losses. The viscous sublayer had a thickness of few millimeters, as depicted in Fig. 3(c); this is the region where viscous forces dominate inertial forces. As a result, turbulent fluctuations are damped and the energy cascade of the turbulent vortices is not present in this region. 500 nm particles cannot diffuse through this viscous sublayer within their time of residence in the tube (approx. 20 s) since their mean displacement due to Brownian motion within that time is of the order of 0.2 mm (see Ref. 27, Table 7.4). As a consequence, particle losses mostly occur in the jet-mixing region and become negligible further downstream. The measured turbulence levels, the mean velocity profile, and the shape of the rms velocity profile indicate that although the Reynolds number is below 2000, the flow is turbulent (plug flow). This ensures that the velocity over the aerosol sampling funnels is nearly uniform and isokinetic sampling is guaranteed with velocity fluctuations of around 10%. Nevertheless, a small uncertainty remains since recent measurements²⁸ of turbulence decay in pipe flows show that vortices, or turbulent puffs, might decay statistically rather than all at the same rate. This effect cannot be captured by the current point measurements, as explained in the first characterization of decaying turbulence in a pipe flow reported by Sibulkin.²⁹ It must be noted, here, that the velocity variation over the sampling cross section would be larger if the flow profile were laminar (parabolic). For instance, for a DUT-OPC with a sampling flow rate of 60 l/min (equal to 1/3 of the total aerosol flow rate), the area of the sampling funnel would correspond to 1/3 of the cross section of the aerosol flow tube. This would result in a velocity difference of about 33% between the particles sampled at the center and close to the periphery of the funnel. This difference would become 16% for OPCs with the standard flow rate of 1 cft/min (28.3 l/min).

The detailed streamwise velocity profile shows that the gradient at the wall is by approximately a factor of two larger in the simulations compared to the measurements. This is probably due to the model's inability to correctly capture the transition between the viscous sublayer and the turbulent boundary layer.

To check for a possible swirling flow, the LDV optics was rotated by 90° around the optical axis. A small swirl was observed in the test section, as illustrated in Fig. 4, with a maximum circumferential velocity of 0.02 m/s, which corresponds to around 15%–20% of the maximum streamwise velocity at the sampling position. By only supplying air to either the three mixing jets (3 times 20 l/min)

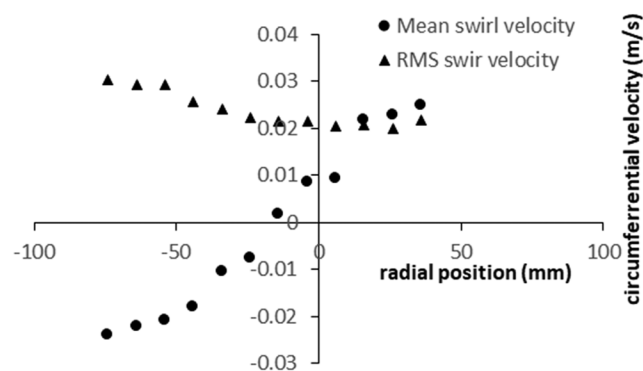


FIG. 4. Measured circumferential mean and rms velocity at streamwise position C.

or the top inlet (120 l/min) and measuring the circumferential velocity, it was found that the swirl resulted solely from the flow of the mixing jets. The reason might be imperfect alignment of the three jets and/or a slight discrepancy in the three flow rates. The flow is controlled by a single mass flow controller (MFC) and is then split and led through three filters, which may exhibit different pressure drops. However, this small swirl does not seem to compromise the performance of the facility in any way. As in the case for the streamwise velocity boundary layer, calculations including swirl show that swirl decays faster in the simulations compared to the measurements, again due to the larger momentum transfer as described above.

B. Particle mixing characteristics

Figure 5 presents the measured particle concentration characteristics for the three downstream positions marked as A', B', and C' in Fig. 1 for the case of the 500 nm and 3000 nm PSL particles and the polydisperse DEHS aerosol. The error bars denote the statistical uncertainty based on the number of sampled particles and amount typically to $\pm 7\%$ and $\pm 0.5\%$ for the PSL and DEHS aerosols, respectively. These uncertainties rely on the assumption of a random Poisson distribution and decrease with increasing particle number concentration. As noted in Subsection III C, the concentration of PSL particles ranged between 2 and 5 cm^{-3} , while the concentration of DEHS particles was larger by three orders of magnitude.

The measurements with 500 nm and 3000 nm PSL particles (positions B' and C') reveal that the aerosol is homogeneous within 4% and 6%, respectively, and imply that the aerosol mixing characteristics are largely independent of the particle size (in the examined size range). The more accurate measurements with DEHS particles indicate a homogeneity within 2.2% at position A' which improves to 1.1% (95% confidence level) at the sampling position C'.

That the mixing is largely independent of the particle size is compatible with the concept of dispersion parameterized using the Stokes number (Stk). The latter is defined as the ratio of the particle relaxation time (τ_p) to the fluid turbulence time scale (τ_f) ($Stk = \frac{\tau_p}{\tau_f}$). The particle relaxation time can be calculated according to equation

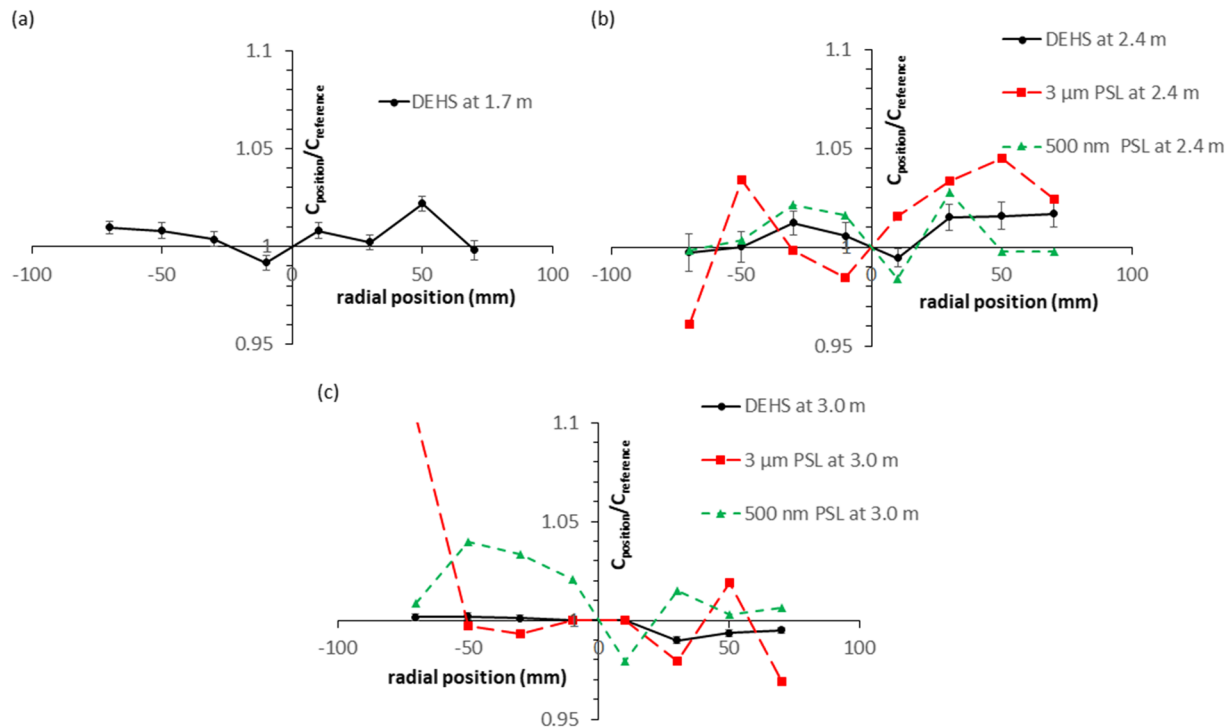


FIG. 5. Measured particle concentration ratios $C_{\text{position}}/C_{\text{reference}}$ for different particle types and sizes at streamwise positions A', B', and C' (see Fig. 1) corresponding to (a) 1.7 m, (b) 2.4 m, and (c) 3.0 m downstream of the aerosol injection position, respectively. $C_{\text{reference}}$ denotes the number concentration along the central axis of the flow tube (radial position = 0 mm), and C_{position} denotes the number concentration at different positions along the diameter of the tube (radial positions of -70 mm, -50 mm, -30 mm, -10 mm, 10 mm, 30 mm, 50 mm, and 70 mm).

$$\tau_p = \frac{\rho d^2}{18\mu}, \quad (1)$$

where ρ designates the particle density, d designates the particle diameter, and μ designates the fluid viscosity. For the current PSL particles (density $\rho = 1.05 \text{ g/cm}^3$), τ_p is found to be $0.9 \mu\text{s}$, $33 \mu\text{s}$, and 0.37 ms for 500 nm, 3 μm , and 10 μm particles, respectively. The fluid turbulence time scale for the vortex at the turbulent jet injection point [see Fig. 2(b)] is estimated to be 350 ms based on the vortex circumference (around 0.65 m) and fluid velocity (downward 3 m/s and upward 0.7 m/s). Downstream in the turbulent pipe, τ_f is calculated to be $\tau_f = \frac{L}{U} = \frac{80 \text{ mm}}{0.2 \frac{\text{m}}{\text{s}}} = 400 \text{ ms}$. Since even the largest vortices are expected to be smaller than the tube diameter, L denotes here the radius (instead of the diameter) of the flow tube and U denotes the bulk velocity of the air. It is known that particle dispersion is equal to that of the fluid for very small Stokes numbers < 0.01 . This is the case for all particle sizes considered in this work with Stk ranging from 2.5×10^{-6} to 0.001. Gravitational settling leads to drift that might reduce dispersion, but the drift is negligible in our case due to the small drift velocity $\leq 3.6 \text{ mm/s}$ (estimated as $\tau_p = 0.37 \text{ ms}$ for 10 μm PSL particles multiplied with the gravitational acceleration $g = 9.81 \text{ m/s}^2$). The gravitational settling is so small that it neither reduces the interaction time with the turbulent vortices nor reduces the residence time in the flow tube considerably.

COMSOL simulations predicted perfect mixing already within the first 0.5–1.0 m downstream of the aerosol injection point. Some uncertainties remain, however, in the model due to the fairly coarse grid around the injection pipes, leading to overestimation of the turbulence and enhanced mixing in the simulations. Additionally, the flow might be dominated by large meandering rather than by fully developed turbulent structures.³¹ Such structures would not mix the particles as turbulent structures, which lead to larger dispersion. We could not, however, confirm the mixing characteristics experimentally so close to the aerosol injection point. This would require inserting in the flow tube sampling pipes longer than 2 m. Since the outlet of the flow tube is situated approximately 1.7 m above the floor, such a test was technically impossible to perform.

V. METROLOGICAL ASSESSMENT AND VALIDATION

A. Uncertainty budget calculation

The particle number concentration at the measurement position can, in principle, be calculated based on the rate of light-scattering incidents and the sampled aerosol volume. In practice, however, particle losses in the tubings connecting the sampling probe to the counter and other artifacts may complicate the measurement. A general expression for the reference particle number concentration, C_{ref} , is given by Eq. (2)

$$c_{\text{ref}} = \frac{N_{\text{average}}(1 - \eta_{\text{noise}})}{QtP} \eta_{\text{hom}}, \quad (2)$$

where N_{average} is the average number of detected light scattering events during repeated measurements of duration t , Q is the aerosol flow, and P is the penetration of particles in the tubings. η_{noise} is associated with the noise signal, and η_{hom} designates the spatial homogeneity of the aerosol across the sampling location. In the following, we discuss briefly each one of the physical parameters entering Eq. (2) and evaluate the associated measurement uncertainty. The results are summarized in Table II. For all quantities, a normal distribution was assumed and the 95% confidence interval was approximated by a coverage factor of 2. Exceptions are the penetration, P , and the homogeneity, η_{hom} , addressed below. The measurement efficiency of the DUT, E_{DUT} , is calculated as $E_{\text{DUT}} = C_{\text{DUT}}/C_{\text{ref}}$, where C_{DUT} is the number concentration measured by the DUT. Due to the much higher aerosol flow and therefore higher particle counts by the DUT, the statistical uncertainties of C_{DUT} are usually small. As a result, the relative uncertainty of E_{DUT} is largely dominated by the uncertainty of C_{ref} . It should be noted, however, that the DUT-OPCs often suffer from large systematic errors, for instance, due to uncertainties in the aerosol flow rate, coincidence or impaction losses, optical misalignment, or false size settings.

1. Uncertainty due to the random particle distribution

In our experiments, every single particle number concentration measurement is performed simultaneously with the DUT-OPC and the reference particle counter and has a duration of 60 s. For a typical concentration of 1 cm^{-3} and the flow rate through the optical cell fixed at 60 ml/min, 60 particles are detected in total by the reference counter within the measurement duration. Such low particle counts ensure that no coincidence losses occur. For calculating the probability of two particles being present in the probe volume, we estimate that the latter corresponds to a circular plane with area of $0.25 \cdot \pi \cdot (0.2 \text{ mm})^2$ (width of aerosol jet = 0.2 mm, see Sec. II C) and a thickness of 0.05 mm (based on Gaussian beam optics). For a mean particle number concentration of 1000 cm^{-3} , this results in an average of $\lambda = 0.00157$ particles being present in the probe volume. The probability of having n particles in this volume is

$$P(n) = \frac{\lambda^n}{n!} e^{-\lambda}. \quad (3)$$

According to Eq. (3), even for an average particle number concentration of 1000 cm^{-3} , the probability of two particles coinciding in the probe volume amounts to be 1.2×10^{-6} and is therefore negligible. For three or more particles coinciding, the probability is even smaller.

Each measurement has a duration of 60 s with the start and end-point being controlled automatically. The uncertainty in the determination of t is therefore negligible. For a typical instrument calibration, the 60 s measurements are repeated 25 times and the average particle number count N_{average} and associated relative standard deviation of the mean are calculated. The latter amounts typically to 5.2% (95% confidence level, see the second row of Table I). Since the particle distribution is random, the theoretical relative standard deviation can be determined using Poisson statistics. Assuming in total $N_{\text{total}} = 60 \cdot 25 = 1500$ particles (for a concentration of 1.0 cm^{-3}),

the relative standard deviation amounts to $\sqrt{N_{\text{total}}}/N_{\text{total}} = 38.7/1500 = 2.6\%$ ($k = 1$) or 5.2% ($k = 2$). A measured standard deviation noticeably larger than the one predicted by theory indicates fluctuations in the aerosol generation (see example in Subsection VI). This simple check can be used as additional quality control of the calibration facility. A Dixon outlier test is also applied, and if more than two of the 25 measurements are found as outliers, all 25 measurements are rejected and the measurement series is repeated.

At particle number concentrations of 100 cm^{-3} , the relative uncertainty in the measured number of particles N reduces to 2.1% (95% confidence level) due to improved statistics. This type of statistical uncertainty could be further reduced by increasing either the measurement duration or the number of measurements.

2. Homogeneity of aerosol sampling and noise

In Fig. 5(c), we presented the cross-stream profile of the mean particle number concentration at the sampling position. From the measurements with DEHS aerosol (only these are discussed here due to their statistical credibility), it can be seen that the concentration fluctuates by $\pm 1.5\%$ with the cross-stream position. The standard deviation of these fluctuations was calculated to be 0.45%. Due to the small number of measurements ($n = 7$), a t -distribution was used, which yields a coverage factor $k \approx 2.4$ for $n - 1 = 6$ degrees of freedom and leads to an uncertainty in the aerosol homogeneity of 1.1% (95% confidence level). In Table II, η_{hom} was therefore set to 1 (perfect homogeneity) with an expanded relative uncertainty of 1.1%.

Bias in the measurement of particle number concentration could also arise due to the detector noise caused, for instance, by random electrons in the photomultiplier. To distinguish between signal peaks and noise spikes, a peak detection algorithm is used and a threshold (trigger) is set manually to define the level above which the data will be collected. Even in the case that noise spikes exceed the trigger level, the peak detection algorithm will assign them to noise based on the signal shape and therefore discard them. In Eq. (1), η_{noise} is therefore set equal to zero.

3. Uncertainty of particle losses and flow rate

Particle losses due to diffusion must also be taken into account on their path from the sampling inlet toward the laser detection area, especially for small PSL particles. The losses depend on particle size, length, and nature of tubings and sampling flow rate and are corrected when calculating c_{ref} using the equation for the penetration P (fraction of exiting to entering particles) provided in the work of Hinds²⁷ [Eq. (7.29)]. A rigorous determination of the uncertainty in P is not feasible since these equations interpolate experimental findings; therefore, an estimated uncertainty of 1% has been assigned according to Ref. 27 (p. 163). Since this is not a statistical uncertainty, the concept of a coverage factor does not apply. The DUT-OPCs, on the contrary, have much higher sampling flow rates ($>0.3 \text{ l/min}$) than the reference optical counter, and therefore, the diffusion losses are negligible. No correction is needed despite the higher Re number (turbulent flow) in the sampling line of the DUT [see Eqs. (7.31) and (7.32) of Ref. 27].

Regarding impaction losses, the vertical design of the setup in conjunction with the isokinetic sampling probes and the straight sampling tubes (practically no bendings) ensures that such losses remain negligible. Only in case the DUT has a horizontal inlet (e.g., CPC) and is to be calibrated with particles larger $1 \mu\text{m}$, the losses

TABLE III. Particle concentrations measured by the DUT-OPC, c_{DUT} , the reference optical particle counter, c_{ref} , and resulting counting efficiency of the DUT-OPC, E_{DUT} . U and s denote expanded uncertainties (95% confidence level).

$d \pm U$ (μm)	$c_{\text{DUT}} \pm s$ (cm^{-3})	$c_{\text{ref}} \pm U$ (cm^{-3})	Expanded relative uncertainty of c_{ref} (%)	$E_{\text{DUT}} \pm U$
0.503 ± 0.13	0.342 ± 0.004	1.45 ± 0.07	4.9	0.24 ± 0.01
1.005 ± 0.024	0.857 ± 0.004	0.88 ± 0.04	5.0	0.97 ± 0.05
5.067 ± 0.045	1.62 ± 0.09	1.98 ± 0.13	6.5	0.82 ± 0.07

due to the unavoidable bend must be calculated based on equations given in Ref. 27 (p. 217). The related uncertainty can be calculated with standard error propagation based on the uncertainty in particle size, aerosol flow, radius of the bend, inclination angle, etc.

The aerosol flow through the reference optical cell is controlled by a calibrated MFC with an uncertainty of 0.3% at 95% confidence (traceable calibration at METAS).

B. Traceability and validation of the facility

The size of the PSL particles used in this study was measured and certified at METAS by means of atomic force microscopy. All mass flow controllers used in the aerosol generation or detection parts of the setup have been calibrated at METAS in a traceable manner (0.3% expanded relative uncertainty). The temperature and pressure in the laboratory are set to $22 \pm 2^\circ\text{C}$ and 950 ± 20 mbar, respectively, and have been calibrated according to the national standards. Finally, the counter and peak-detection algorithm of the reference particle counter have been calibrated with respect to the national standard of frequency by generating pulses of known frequency, duration, and line shape (single and double peaks). It should be also noted that counting and using the unit 1 is an integral part of the SI definition.³²

Furthermore, a round-robin test involving the IAG¹⁷ and our setup, performed in the size range $d_{\text{PSL}} = 500 \text{ nm} - 5 \mu\text{m}$ and at particle number concentrations $\leq 1 \text{ cm}^{-3}$, has yielded an agreement typically within 4% in the efficiency calibration of three different DUT-OPCs.³³ Our reference optical particle counter has been further validated at $d_{\text{PSL}} = 100 \text{ nm}$ with a CPC, which had been previously calibrated in a traceable manner with an electrometer according to ISO 27891. Measurements were performed at concentrations of a few hundred particles per cm^3 , and the results agreed within 3%.

VI. EXAMPLE OF OPC CALIBRATION

In this section, we provide a real example of an OPC calibration performed at a number concentration of approximately 1 cm^{-3} . The results are summarized in Table III. The particles with a nominal diameter of $5 \mu\text{m}$ were generated with the fluidized bed generator (see Subsection II A). As discussed in Subsection V A, due to fluctuations in the aerosol generation when using the fluidized bed system, the uncertainty of c_{ref} at $5 \mu\text{m}$ is larger than that at $0.5 \mu\text{m}$ and $1.0 \mu\text{m}$. The increase in the statistical uncertainty (standard deviation of the mean) s of c_{DUT} is even more pronounced. This results in larger uncertainties in the determination of the counting efficiency of the DUT-OPC, E_{DUT} , at $5 \mu\text{m}$.

VII. SUMMARY AND CONCLUSIONS

This report describes a custom-made facility to calibrate optical and condensation particle counters, light scattering aerosol spectrometers, and aerodynamic particle sizers. PSL particles in the size range $100 \text{ nm} - 10 \mu\text{m}$ are injected in a flow tube and mixed by three air-jets. At the bottom of the flow tube, the particles are sampled isokinetically using probe funnels with sizes adjusted to the flow rate of the DUT and the reference particle counter, respectively. With this facility, aerosol number concentrations in the range 0.5 cm^{-3} up to a few thousand per cm^3 can be generated and measured without coincidence losses.

A detailed uncertainty budget was established taking into account statistical uncertainties, the aerosol spatial inhomogeneity at the two sampling funnels, and possible particle losses due to diffusion and impaction in the sampling tubes. At a concentration of 1 cm^{-3} , the uncertainty in the measurement of the particle number concentration was determined to be 5.4% (95% confidence level), dominated by the random distribution of particles and the low counting statistics of the reference particle counter.

The flow in the facility was characterized in terms of velocity fields, obtained by measurements and simulations. It was found that the flow, despite a Reynolds number of around 1800, remained turbulent with a nearly flat velocity profile in the sampling zone. Moreover, the mixing characteristics of the injected PSL particles were studied experimentally. At the outlet of the facility, the aerosol homogeneity across the flow tube diameter was determined to be within 1.1%.

Future work will focus on improving the design of this facility by reducing the dimensions of the flow tube. In a second step, the performance of the facility will be evaluated with aerosols from different sources, such as soot, salt, and dust with the aim to generate multicomponent aerosols that mimic the properties of outdoor air.

ACKNOWLEDGMENTS

The study of the flow characteristics (LDV measurements and CFD simulations presented in Subsections III A and III B) and the evolution of particle mixing characteristics (Subsection III C) were carried out in the context of the 16ENV07 Aeromet Project funded by the European Union through the European Metrology Programme for Innovation and Research (EMPIR). EMPIR is jointly funded by the EMPIR participating countries within EURAMET and the European Union. METAS was supported by the Swiss State Secretariat for Education, Research and Innovation (SERI) under Contract No. 17.00112. The opinions expressed and arguments

employed herein do not necessarily reflect the official views of the Swiss Government.

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